Robert Andrew Conrad

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Experience (Past 5 Years)

- **Acre Security** Ottawa,ON
  Senior Software Developer, L2 *Feb 2026 - Present*
  - **Full Stack Development**: Developed full stack products across C#.NET, Flutter & JavaScript leveraging MQTT, MongoDB and SQL systems for cloud synchronized access of access control hardware and video information
  - **Integrations**: Design and Develop 3rd Party integrations for Access Control solutions including Video Management System, Hardware solutions, Mobile Credentials and Biometrics from technical design to market support
  - **AI Builder**: Trained to utilize and apply safe AI development strategies including agentic programming and auditable AI assisted tooling
- **Face 4 Systems** Ottawa,ON
  Software Engineering *4.5 Years. July 2021 - January 2026*
  - **Mobile & Web**: Developed multiple cross platform mobile apps in Flutter/React/C++/C# MAUI.NET to perform Face Recognition enrollment and access control use cases in both stationary and "On The Go" situations. Process includes UX and architecture design, testing and Continuous Integration(CI)
  - **Backend**: Designed REST API Clients in C#.Net, Flutter and JavaScript to perform image analysis, system controls and large scale data migration. Integrated 3rd Party interfaces and deployed in a secure and easily expandable environment
  - **DevOps**: Created workflows to support deployment of the application in AWS for customer use using containerization and scripts created in Bash, Powershell, Javascript and Golang
  - **Design and Support**: Develop applications and services to support facial recognition and image quality for access control and identity use cases working with customer from initial design to deployment and maintenance.
- **Quantropi** Ottawa,ON
  Engineering Intern *May 2020 - Dec 2020*
  - **DevOps**: Developed a quality assurance and CI process for dynamic C++ and Python SDK delivery using CMake and Bash
  - **QA**: Created an automation script that randomized inputs that could integrating into existing CMake and xUnit frameworks
- **Ciena** Ottawa,ON
  Software Developer, Coop *Sept 2019 - Dec 2019*
  - **Distributed Apps**: Developed a cross system micro service in Go for logging, data export and REST API configuration
- **Pleora Technology** Ottawa,ON
  Embedded Software Developer *Jan 2019 - April 2019*
  - **Firmware and Licencing Tools**: Developed new tools to recognize new camera specifications and apply licencing in C# and C++
- **Nordiasoft** Gatineau,QC
  Mobile Application Developer *May 2018 - August 2018*
  - **Android**: Created a Android app in Java that turned a phone into a radio input and controller using a SDR

Programming Skills

- **Programming Language** C# , Flutter/Dart, C/C++, Golang, JavaScript/Typescript , Python
- **Frameworks & Tools** .NET (C# MAUI, ASP.NET .NET), SQL (SQLite, PostgreSQL, Microsoft), MongoDB , Claude Code, GitHub Copilot
- **DevOps** VCS (Git,Mercurial,CVS), CI (Jenkins, GitHub Actions), Docker/Podman, VMWare , AWS (EC2,Route 53), CMake

Education

- **University of Ottawa** Ottawa,ON
  Bachelor of Applied Science,Computer Engineering (Management & Coop Options) Cum laude *2016-2021*
  - **Topics**: Programming, Algorithms, Systems Design, Database, AI, Networking etc
  - **Extra Curricular**: IEEE uOttawa Branch Treasurer/Chair, uOttaHack Founding Team 2017/19,Ottabotics Software Team Lead, ESSCO Board Chair, Endowment Fund Committee 2018-22

Projects

- **uOttawa Roboboat** University of Ottawa
  Software Co-lead *September 2017 - December 2018*
  - **Architecture**: Designed systems architecture for multi competition autonomous surface vehicle in ROS, Python and C++. Primary focus on control integration and core behavior systems
  - **Algorithms**: Worked alongside masters student to determine algorithms to solve each navigation challenge, including RA* and decision tree

- **Capstone Research: Retirement Hygiene Management** University of Ottawa
  Team Lead *January-December 2020*
  - **Design and Proposal**: Developed a prototype system for local medical technology engineer to monitor water use of an individual in a private and none invasive way to allow longer self autonomy per the Emerging Technologies Report of 2019
  - **Hardware**: Created Hardware side REST API Call to transit Data over REST API via Python and Go with a option Camera Module
  - **Architecture**: Designed systems architecture including hardware design and network architecture for local hardware to communicate with Healthcare Agent Data Portal